

Outcomes of liver transplantation in patients 60 years and older: a multivariate analysis using penalized Cox regression

Introduction:

To enhance our understanding of variables that influence transplantation outcomes, in this study, we focus on a subset of the population of donors aged 60 and above given the increasing prevalence of liver transplants in the elderly. Utilizing a multivariate analysis, we identified those variables with an impact on 1-year, 2-year, and 3-year survival.

Method:

Our study investigated an American cohort of 21,625 individuals aged 60 and older, sourced from the Scientific Registry of Transplant Recipients (SRTR). Survival rates at one, two, and three years' post-transplant were analysed. Transplants from living donors were excluded, and missing data were imputed using mean values for each variable (except survival times, which were addressed using a censoring indicator mechanism). Penalized Cox regression was employed to identify variables significantly correlated with survival time for each duration, with selection criteria optimized through cross-validation of the model's concordance index.

Results:

Of the twenty variables significantly correlated to survival in each group, only five were common across all three survival groups: the MELD score, home state, CMV infectious status, antiretroviral therapy treatment, and preservation solution. Previous transplantation appeared to be a risk factor for the 1-year survival group, while recipient age emerged as a risk factor for the 2-year survival group. Malignancy status showed a stronger correlation with the 3-year survival group.

Conclusion:

This study offers novel insights into the risk factors affecting mid-term survival among liver transplant recipients aged 60 and above. The disparate sets of significant variables observed across the three survival periods underscore the complexity of developing a unified predictive model. Nonetheless, these preliminary findings lay a foundation for the development of a predictive scoring system to refine donor-recipient matching and optimize survival outcomes.